

5.3 Range and measures of centre



In the previous sections you have seen how to summarise data in the form of a frequency table and display data using graphs. Key summary statistics also allow us to describe the data using a single numerical value. The mean ('average'), for example, may be used to describe a student's performance over a series of tests. The median (middle value when data is ordered smallest to largest) is often used when describing the house prices in a suburb. These are termed *measures of centre*. Providing information about the spread of the data is the range, which measures the difference between the maximum and minimum values.



► Let's Start: Mean, median or mode?

The following data represents the number of goals scored by Ellie in each game of a 9-game netball season.

24 18 25 16 3 23 27 19 25

It is known that the figures below represent, in some order, the mean, median and mode.

25 20 23

- Without doing any calculations, can you suggest which statistic is which? Explain.
- From the data, what gives an indication that the mean ('average') will be less than the median (middle value)?
- Describe how you would calculate the mean, median and mode from the data values.

- The **mean** (or average) is calculated by summing all the data values and dividing by the total number of values.

$$\text{Mean } (\bar{x}) = \frac{\text{sum of all data values}}{\text{number of data values}}$$

- The mean is affected by extreme values in the data.

- The **mode** is the most commonly occurring value in the data set.

- A data set can have two modes (called **bimodal**) or no unique mode at all.

- The **median** is the middle value of a data set when the data is arranged in order.

- If the data set has an even number of values the median is the average of the two middle values.

For example,

2 3 ⑥ 8 12

Median = 6

4 7 8 10 13 17

$$\text{Median} = \frac{8 + 10}{2} = 9$$

- The mean is affected by extreme values in the data.

Mean An average value calculated by dividing the total of a set of numbers by the number of values

Mode The score that appears most often in a set of numbers

Bimodal When a set of data has two modes

Median The middle score when all the numbers in a set are arranged in order

- The **range** is a measure of how spread out the data is.
– Range = maximum value – minimum value

Range The difference between the highest and lowest numbers in a set

Exercise 5C

Understanding

- Use the words from the list below to fill in the missing word in these sentences.
mean, median, mode, bimodal, range
 - The _____ is the most frequently occurring value in a data set.
 - Dividing the sum of all the data values by the total number of values gives the _____.
 - The middle value of a data set ordered from smallest to largest is the _____.
 - A data set with two most common values is _____.
 - A data set has a maximum value of 7 and a minimum value of 2. The _____ is 5.
- Calculate the following.
 - $\frac{1+4+5+8+2}{5}$
 - $\frac{3+7+6+2}{4}$
 - $\frac{3.1+2.3+6.4+1.7+2.5}{5}$
- Circle the middle value(s) of these ordered data sets.
 - 2 4 6 7 8 10 11
 - 6 9 10 14 17 20
- Michael drinks the following number of cups of coffee each day in a week:
4 5 3 6 4 3 3
 - How many cups of coffee does he drink in the week (sum of the data values)?
 - How many days are in the week (total number of data values)?
 - What is the mean number of cups of coffee Michael drinks each day (part **a** ÷ part **b**)?

Recall that an even number of data values will have two middle values.



Fluency

Example 6 Finding the mean, mode and range

For the following data sets, find:

- the mean
 - the mode
 - the range
- 2, 4, 5, 8, 8
 - 3, 15, 12, 9, 12, 15, 6, 8

Solution

$$\begin{aligned} \text{a i Mean} &= \frac{2+4+5+8+8}{5} \\ &= \frac{27}{5} \\ &= 5.4 \end{aligned}$$

ii The mode is 8.

$$\begin{aligned} \text{iii Range} &= 8 - 2 \\ &= 6 \end{aligned}$$

Explanation

$$\text{Mean} = \frac{\text{sum of all data values}}{\text{number of data values}}$$

Add all the data and divide by the number of values (5).

The mode is the most common value in the data.

$$\text{Range} = \text{maximum value} - \text{minimum value}$$

$$\begin{aligned}\text{b i Mean} &= \frac{3 + 15 + 12 + 9 + 12 + 15 + 6 + 8}{8} \\ &= \frac{80}{8} \\ &= 10\end{aligned}$$

ii There are two modes, 12 and 15.

$$\begin{aligned}\text{iii Range} &= 15 - 3 \\ &= 12\end{aligned}$$

$$\text{Mean} = \frac{\text{sum of all data values}}{\text{number of data values}}$$

Add all the data and divide by the number of values (8).

The data is bimodal: 12 and 15 are the most common data values.

$$\text{Range} = \text{maximum value} - \text{minimum value}$$



5 For each of the following data sets, find:

i the mean ii the mode iii the range

a 2 4 5 8 8

b 5 8 10 15 20 12 10 50

c 55 70 75 50 90 85 50 65 90

d 27 30 28 29 24 12

e 2.0 1.9 2.7 2.9 2.6 1.9 2.7 1.9

f 1.7 1.2 1.4 1.6 2.4 1.3

Recall:

$$\text{Mean} = \frac{\text{sum of data values}}{\text{number of data values}}$$

Mode is the most common value

$$\text{Range} = \text{maximum} - \text{minimum}$$



Example 7 Finding the median

Find the median of each data set.

a 4, 7, 12, 2, 9, 15, 1

b 16, 20, 8, 5, 21, 14

Solution

a 1 2 4 ⑦ 9 12 15

$$\text{Median} = 7$$

b 5 8 14 16 20 21

$$\begin{aligned}\text{Median} &= \frac{14 + 16}{2} \\ &= 15\end{aligned}$$

Explanation

The data must first be ordered from smallest to largest. The median is the middle value.

For an odd number of data values, there will be one middle value.

Order the data from smallest to largest.

For an even number of data values, there will be two middle values.

The median is the average of these two values (the value halfway between the two middle numbers).

6 Find the median of each data set.

a 1 4 7 8 12

b 1 2 2 4 4 7 9

c 11 13 6 10 14 13 11

d 62 77 56 78 64 73 79 75 77

e 2 4 4 5 6 8 8 10 12 22

f 1 2 2 3 7 12 12 18

g 30 36 31 38 27 40

h 2.4 2.0 3.2 2.8 3.5 3.1 3.7 3.9

First make sure that the data is in order. For two middle values, find their average.



5C

- 7 Nine people watch the following number of hours of television on a weekend.

4 4 6 6 6 8 9 9 11

- a Find the mean number of hours of television watched.
- b Find the median number of hours of television watched.
- c Find the range of the television hours watched.
- d What is the mode number of hours of television watched?

- 8 Eight students compare the amount of pocket money they receive. The data is as follows.

\$12 \$15 \$12 \$24 \$20 \$8 \$50 \$25

- a Find the range of pocket money received.
- b Find the median amount of pocket money.
- c Find the mean amount of pocket money.
- d Why is the mean larger than the median?

Problem-solving and Reasoning



Example 8 Calculating summary statistics from a stem-and-leaf plot

For the data in this stem and leaf plot, find:

- a the range
- b the mode
- c the mean
- d the median

Stem	Leaf
2	5 8
3	1 2 2 2 6
4	0 3 3
5	2 6
5 2 means 52	

Solution

- a Minimum value = 25
Maximum value = 56
Range = $56 - 25$
= 31

- b Mode = 32

Explanation

In an ordered stem-and-leaf plot the first data item is the minimum and the last is the maximum. Use the key '5 | 2 means 52' to see how to put the stem and leaf together.

Range = maximum value – minimum value

The mode is the most common value. The leaf 2 appears three times with the stem 3.

$$\begin{aligned} \text{c Mean} &= \frac{24 + 28 + 31 + 32 + 32 + 32 + 36 + 40 + 43 + 43 + 52 + 55}{12} \\ &= \frac{450}{12} \\ &= 37.5 \end{aligned}$$

$$\begin{aligned} \text{d Median} &= \frac{32 + 36}{2} \\ &= 34 \end{aligned}$$

Form each data value from the graph and add them all together. Then divide by the number of data values in the stem-and-leaf plot.

There is an even number of data values: 12. The median will be the average of the middle two values (the 6th and 7th data values).



9 For the data in these stem-and-leaf plots, find:

i the range

iii the mean (rounded to one decimal place)

ii the mode

iv the median

a

Stem	Leaf
2	1 3 7
3	2 8 9 9
4	4 6

3|2 means 32

b

Stem	Leaf
0	4 4
1	0 2 5 9
2	1 7 8
3	2

2|7 means 27

c

Stem	Leaf
10	1 2 4
11	2 6
12	5

11|6 means 116

d

Stem	Leaf
3	0 0 5
4	2 7
5	1 3 3
6	0 2

3|2 means 3.2

Use the key to see how the stem and leaf go together.



10 This back-to-back stem-and-leaf plot shows the results of two students, Hugh and Mark, on their end-of-year examination in each subject.

a For each student, find:

i the mean

ii the median

iii the range

b Compare the performance of the two students using your answers to part **a**.

Hugh leaf	Stem	Mark leaf
8 8 5	6	4
7 3	7	4 7
5 4 2 1 1	8	2 4 6 8
	9	2 4 5

7|4 means 74%

11 A real estate agent recorded the following amounts for the sale of five houses:

\$120 000 \$210 000 \$280 000 \$370 000 \$1 700 000

The mean is \$536 000 and the median is \$280 000.

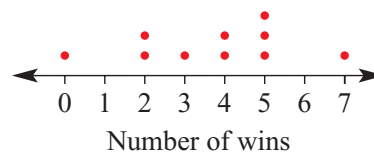
Which is a better measure of the centre of the five house prices: the mean or the median?

Give a reason.

5C

12 This dot plot shows the number of wins recorded by a school sports team in the last 10 eight-game seasons.

- a** What was the median number of wins?
b What was the mean number of wins?
c The following season, the team records 3 wins. What effect will this have (increase/decrease/no change) on the:
i median? **ii** mean?



13 Catherine achieves the following scores on her first four maths tests: 64 70 72 74

- a** What is her mean mark from the maths tests?
b In the fifth and final test, Catherine is hoping to raise her mean mark to 73. What mark does she need on the last test to achieve this?

A mean of 73 from 5 tests will need a five-test total of 73×5 .



Moving run average

14 A moving average is determined by calculating the average of all data values up to a particular time or place in the data set.

Consider a batsman in cricket with the following runs scored from 10 completed innings.

Innings	1	2	3	4	5	6	7	8	9	10
Score	26	38	5	10	52	103	75	21	33	0
Moving average	26	32								

In the table, 26 is the average after 1 innings and 32 is the average after 2 innings.

- a** Complete the table by calculating the moving average for innings 3–10. Round to the nearest whole number where required.
b Plot the score and moving averages for the batsman on the same set of axes, with the innings number on the horizontal axis. Join the points to form two line graphs.
c Describe the behaviour of the:
i score graph
ii moving average graph
d Describe the main difference in the behaviour of the two graphs. Give reasons.